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Enhancement of surfactant biodegradation with an anaerobic membrane bioreactor by introducing microaeration

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HIGHLIGHTS

- Microaeration effectively enhanced biodegradation of anionic surfactants.
- The swift degradation followed the kinetic characteristics of zero order reaction.
- The biodegradation processes were clearly illustrated by 3D-EEM and SEC results.
- Specific genera of bacteria involving with biodegradation processes were found.

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ABSTRACT

In controlled ecological life support system (CELSS), 100% water recycle efficiency is needed. Both water and nutrients in wastewater needs reclaiming. In an anaerobic membrane bioreactor (AnMBR) designed to hydrolyze organic matters and convert organic nitrogen, accumulation of anionic surfactants and organic compounds was observed in the supernatant. To solve the problem, microaeration which had rarely been reported to enhance surfactants biodegradation in anaerobic process was introduced and it was proved to be an effective approach that resulted in concentrations of surfactants and soluble chemical oxygen demand (SCOD) in the supernatant decreasing from 9000 mg/L and 40000 mg/L to 2000 mg/L and 10000 mg/L, respectively. And the degradations followed the kinetic characteristics of zero order reaction with R^2 values of 0.9472 and 0.949. Three-dimensional excitation emission matrix (3D-EEM) analysis revealed that some activities of microbes were activated and enhanced by microaeration, which resulted in the disintegration of aggregates of large molecules, indicated by the size exclusion chromatography (SEC). After the introduction of microaeration, the emerging genera *Aquamicrobium*, *Flaviflexus*, *Pseudomonas* and *Thiopseudomonas* in the microbial community might be responsible for the effective biodegradation of the surfactants.

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1. Introduction

Controlled ecological life support system (CELSS) is an artificial closed ecosystem built for long-term space exploration which provides humans (astronauts) with enough breathable air, clean water, essential food and safe working-living places (Dai et al., 2018). In the CELSS, urine and greywater need properly treating and a 100% water recovery rate is essential to ensure the material closure of the whole system. And all the waste treatment systems have to feature high performance, small foot prints, and as little consumables and by-products (such as gas emission and excess sludge) as possible. In 1990s, National Aeronautics and Space Administration (NASA) made an estimation of 20–30 L/(d·person) wastewater in the CELSS (Wieland, 1994; Wydeven and Golub,

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Nomenclature

3D-EEM	three-dimensional excitation emission matrix
AMW	apparent molecular weight
AnMBR	anaerobic membrane bioreactor
AU	arbitrary units
CELSS	controlled ecological life support system
CMC	critical micelle concentration
DOM	dissolved organic matters
Em	emission
Ex	excitation
FRI	fluorescence regional integration
MBAS	Methylene blue active substance
NASA	National Aeronautics and Space Administration
SEC	size exclusion chromatography
SLES	sodium lauryl polyoxyethylene ether sulfate
SLS	sodium lauryl sulfate

1990). And results from the recent 4-person-180-day integrated experiment inside a CELSS conducted in China which employed a combined AnMBR-MBR system (our previous work) in wastewater treatment and reuse process indicated that an average of 51 L wastewater was produced per day (Li and Zuo, 2017). Compared with a water consumption of 75–220 L/(d·person) for city's residential use (Ministry of Construction of the People's Republic of China, 2002), the much lower water consumption in the CELSS results in higher concentration of surfactants contained in personal care products and detergents in the wastewater. In NASA's advanced life support project, greywater recycling was studied through the use of a fixed-film bioreactor, with higher than 600 mg/L surfactant (Igepon TC-42) in the influent (Levine et al., 2000).

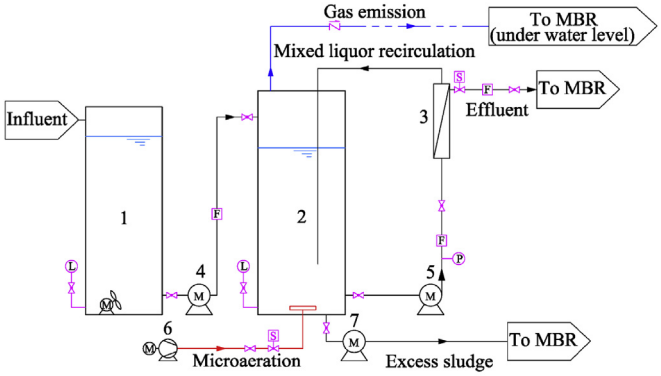
Anionic surfactants, as core components of detergent and cosmetic products formulations, contribute significantly to the pollution profile of sewage and wastewaters of all kinds. In the CELSS, purified wastewater is usually recycled into the hydroponic system. Surfactants have been considered to have phytotoxic effects (Bubenheim et al., 1997) and their accumulation in the hydroponic system may be barely avoided, so they have to be eliminated as possible. Surfactants at high concentrations in wastewater can be eliminated effectively by coagulation or foam separation (Beltrán-Heredia et al., 2012; Srinet et al., 2017). But such processes produce large quantity of solid waste, making them not suitable for the CELSS. Though surfactants are degradable or readily degradable under aerobic condition (Lechuga et al., 2016), they can give rise to some problems such as foaming and long retention times in the wastewater treatment (McAvoy et al., 1993). Thus treatment of undiluted wastewaters under anaerobic conditions may partially surmount this difficulty (Shcherbakova et al., 1999). However, surfactants have been shown to inhibit the anaerobic bioprocess (Feitkenhauer and Meyer, 2002) and results of some studies (J. Krueger et al., 1998; Scott and Jones, 2000) have demonstrated the persistence of surfactants under anaerobic conditions. Microaeration i.e. some amount of oxygen is introduced to a basically anaerobic biochemical process (Botheju and Bakke, 2011) has been applied to enhance the anaerobic bioprocess. Some studies showed that under proper experimental conditions, mainly an appropriate amount of air or oxygen introduced, and enhancement of hydrolysis (Wu et al., 2015), sulfide oxidation (van der Zee et al., 2007), elevation of methane yield (Tartakovsky et al., 2011), and degradation of refractory and toxic organic compounds (Zitomer and Shrout, 1998) have been achieved. Whereas whether or not effective biodegradation of surfactants can occur under microaeration condition has remained unknown during anaerobic treatment.

In this study, an AnMBR in the combined AnMBR-MBR system for wastewater treatment and reuse was used to treat synthetic CELSS wastewater. The AnMBR was controlled under hydrolysis-acidogenic stage aiming at realizing ammonification, primary breakdown of complex organic compounds and in the meanwhile avoiding large amount of biogas production according to requirements of the CELSS. Compared to some high strength wastewater such as that is produced by food manufacturing process, CELSS wastewater has much less organic compounds, which means its biochemical methane potential is correspondingly much lower. And what's more, methanogenenic process needs longer hydraulic retention time (HRT) that demands bigger footprint, and the systematic complexity of CELSS will be higher for utilizing methane. So, maybe it's better to employ a hydrolysis-acidogenic process as the initial stage of the CELSS wastewater treatment. Therefore, the purpose of this study was to investigate performance of the AnMBR with the introduced microaeration, especially on anionic surfactants biodegradation, including characteristics of surfactants degradation, transformation of dissolved organic matters (DOM) and evolution of microbial community, in order to provide technical support for the CELSS wastewater treatment and reuse.

2. Materials and methods

2.1. Description of the AnMBR

The AnMBR comprised an anaerobic bioreactor and two pressure-driven external cross-flow tubular membrane modules (MO P1U(0.5 m)_I8, Beijing Tri-high Membrane Technology CO., Ltd, PVDF, 0.03 μm pore size, 0.011 m^2 filtering area) in series (Fig. 1). The cross-flow rate of the mixed liquor was kept above 2.5 m/s by a centrifugal pump (CRN 1s-2, Grundfos) to control membrane fouling, and the permeate flux was set at around 27 L/(m^2·h). The anaerobic bioreactor was a continuous stirred-tank reactor. It was made of polymethyl methacrylate and had a diameter of 190 mm and a height of 700 mm. And it was sealed with a cover plate to prevent uncontrolled gas emission. A buffer tank also made of polymethyl methacrylate with a maximal working volume of 78 L was used to store the synthetic wastewater. An air compressor was used to supply limited amount of air into the anaerobic bioreactor. To avoid severe foaming problem, the air was introduced through a plastic tube with an inner diameter of 3 mm.



1 Buffer tank, 2 Anaerobic bioreactor, 3 Tubular membrane module, 4 Inlet pump, 5 Mixed liquor circulation pump, 6 Blower, 7 Sludge pump

Fig. 1. Schematic diagram of the AnMBR.

Excess sludge was regularly discharged with a sludge pump according to a specific sludge retention time (SRT). The system ran automatically under control of a programmable logic controller (PLC) (S7-200, Siemens).

2.2. Synthetic wastewater

Synthetic wastewater used in this study was made up of two parts, i.e. synthetic urine and greywater. The recipe of synthetic urine was according to Chutipongtanate and Thongboonkerd (2010) and Verostko et al. (2004). Amounts of detergent constituents such as toothpaste, body lotion and liquid soap etc. in the synthetic greywater were estimated based on a 4-person crew that produced about 50 L wastewater altogether per day. And other organic compounds were added according to Chu et al. (2005). The surfactants in the synthetic wastewater were sodium lauryl sulfate (SLS) and sodium lauryl polyoxyethylene ether sulfate (SLES) which were two typical sorts of anionic surfactant used in detergents. The major water quality parameters were as follows: COD 1200–1400 mg/L, SCOD 1000–1300 mg/L, NH_4^+–N 70–80 mg/L, TN 700–800 mg/L and anionic surfactants 110–120 mg/L.

2.3. Operation and sampling

The AnMBR was seeded with concentrated excess sludge from a municipal wastewater plant with an anaerobic/anoxic/oxic (A^2/O) process at capacity of 550,000 m^3/d in Beijing. Before this study, the AnMBR had run for 5 months treating this synthetic wastewater. During the first 60 days of this study, the mixed liquor suspended solids (MLSS) in the reactor was maintained at 29.8 ± 2.95 g/L and decreased to 15.9 ± 2.85 g/L afterwards because some sludge was purged out of the AnMBR for some other experiments. Sample of sludge (100 mL) was taken once a week and 40 mL sludge was discharged every day, resulting in a SRT of 243 d. The HRT was set at 20 h. As shown in Table 1, during the first 68 days, no air was introduced into the AnMBR and in the following periods different amounts of limited air were introduced with a fishbowl air compressor with power capacity of 20 W. An ORP of −200 mV or lower was maintained throughout the whole experimental period. Although the AnMBR ran at ambient temperature, the temperature of the mixed liquor was maintained between 33 and 38 °C due to the continuous circulation. Samples of influent and permeate were taken at least twice a week. Supernatants were collected by centrifuging mixed liquor at 10000 rpm for 10 min. Samples of influent, supernatant and permeate were kept at −4 °C and samples of mixed liquor were kept at −20 °C before analysis.

2.4. Analytical methods

2.4.1. Water quality

Ammonia and total nitrogen (TN) in the influent and AnMBR permeate were analyzed according to the Standard Methods (APHA, 2005). Chemical oxygen demand (COD) was measured using HACH 8000 methods with a DR 2800 spectrometer (HACH, USA). SCOD was measured with the filtrate samples through 0.22 μm membrane (SCAA-1202, Anpel, China). Methylene blue active substances (MBAS) assay was carried out to determine anionic surfactants concentration using HACH Crystal Violet Method (Method 8028) and the result was expressed as mg MBAS/L. MLSS and mixed liquor volatile suspended solids (MLVSS) were determined at 104 °C (4 h) and 600 °C (2 h), respectively.

Table 1
The major operation parameters of the AnMBR.

Stage	Time, d	Microaeration strategy
I	1~68	none
II	69~140	A 0.5 L/min flow rate for 6s of every 2 min
III	141~225	A 0.3 L/min flow rate for 6s of every 3 min

2.4.2. Measurement of DOM

Concentrations of six volatile fatty acids (VFAs) namely acetic acid, propionic acid, butyric acid, *iso*-butyric acid, valeric acid and *iso*-valeric acid were analyzed by a gas chromatograph (Shimadzu GC-2010 Plus, Japan) equipped with a flame ionization detector and stabilwax-DA column (0.32 mm × 25 m × 0.25 μm) as described previously (Zhang et al., 2016). The total concentration of VFAs was described as mg COD/L.

A Varian Eclipse fluorescence spectrophotometer (F-7000, Hitachi, Japan) was used to acquire 3D-EEM fluorescence spectra of the filtered supernatants in different experimental stages. The spectrometer used a xenon excitation source and displayed a maximum emission intensity of 10000 arbitrary units (AU). EEM spectra were gathered with scanning emission (Em) spectra from 220 to 550 nm at 5 nm increments by varying the excitation (Ex) wavelength from 200 to 400 nm at 5 nm increments. The spectra were recorded at a scan rate of 12000 nm/min, using excitation and emission slit bandwidths of 5 nm (He et al., 2017). Three Ex/Em regions (Region I: Ex 200–250 nm and Em 280–330 nm; Region II: Ex 200–250 nm and Em 330–380 nm; Region III: Ex 250–380 nm and Em 280–380 nm) of the EEM spectra were analyzed using the fluorescence regional integration (FRI) method (Chen et al., 2003).

The percent fluorescence response ($P_{i,n}$) of samples in different stages were calculated as follows (Chen et al., 2003):

$$P_{i,n} = \Phi_{i,n}/\Phi_{T,n} \times 100\% \quad (1)$$

$$\Phi_{i,n} = MF_i\Phi_i \quad (2)$$

$$\Phi_{T,n} = \sum_{i=1}^{3}\Phi_{i,n} \quad (3)$$

$$\Phi_i = \sum_{Ex}\sum_{Em} I(\lambda_{Ex}\lambda_{Em})\Delta\lambda_{Ex}\Delta\lambda_{Em} \quad (4)$$

in which Φ_i is the volume beneath region "*i*"; I (λ_{Ex}, λ_{Em}) is the fluorescence intensity at each *Ex*-*Em* wavelength pair; $\Delta\lambda_{Ex}$ is the *Ex* wavelength interval (taken as 5 nm); $\Delta\lambda_{Em}$ is the Em wavelength interval (taken as 5 nm); MF_i is the inverse of the fractional projected excitation-emission area; $\Phi_{i,n}$ and $\Phi_{T,n}$ are normalized *Ex*-*Em* area volumes.

The apparent molecular weight (AMW) distribution of the effluent compounds in different experimental stages were analyzed by SEC with a high performance liquid chromatography system (Breeze 1525, Waters Co, USA) using a UV detector (Waters 2487 Dual λ Absorbance Detector) at 254 nm. A TSKgel $G3000SW_{XL}$ type gel column (Tosoh Corporation, Japan) was used with 5 mM sodium phosphate buffer (including Na_2HPO_4 and NaH_2PO_4) and 10 mM sodium chloride solution as eluent at a flow rate of 0.8 mL/min.

2.4.3. Microbial community analysis

Microbial communities of sludge samples of different experimental stages as well as the inoculum were analyzed. 1 mL of each sludge sample was centrifuged at 10,000 rpm for 10 min, and the pellet was used for DNA extraction using FASTDNA Spin Kit for Soil (MP Biomedicals, USA) in triplicate according to manufacturer's instructions, and the resulting extracts were composited to average

out bias in sampling and extraction. Quality and concentration of the extracted DNA were determined through 1% agarose gel electrophoresis and NanoDrop ND-1000 (NanoDrop, USA), respectively.

PCR primers 515F/806R targeting the bacteria and archaea 16S V4 region were selected for the microbial community structure analysis. The reverse primer contains a 6-bp error-correcting barcode unique to each sample. The barcode was permuted for each sample and permitted the identification of individual samples within a mixture in a single Illumina MiSeq sequencing run. DNA was amplified in triplicate for each sample following a protocol described previously (Zhang et al., 2016). PCR amplicons were further purified using a DNA purification kit (BioFlux, Japan), and the concentrations were determined by spectrometry using a QuantiFluor™-ST (Promega, USA). Amplicons from different samples were then mixed to achieve equal mass concentrations in the final mixture, which were sent out to Sangon Co., Ltd. in Shanghai for small-fragment library construction and pair-end sequencing using the Illumina MiSeq sequencing system (Illumina, USA).

Sequencing reads were assigned to each sample according to the unique 6-bp barcode for each sample. Pairs of reads from the original DNA fragments were merged using FLASH and then were filtered using QIIME quality filters. PCR chimeras were filtered out using UCHIME, and the rest sequences were normalized to fairly compare the samples in the same sequencing depth. The taxonomic classification of the sequences in each sample was carried out individually using RDP Classifier, and the sequences of different taxonomy levels were assigned at the bootstrap cutoff of 50%, as suggested by the RDP. In addition, diversity and richness indices were calculated using the relevant RDP pipeline modules.

3. Results and discussion

3.1. Performance of the AnMBR

Throughout the whole experimental period, the influent COD concentration was steadily around 1300 mg/L, and it could be divided into two sub-periods according to the COD removal efficiency. As shown in Fig. 2(a), in Stage I without microaeration the effluent COD concentration was 800–1000 mg/L, resulting in COD removal efficiency between 25 and 40%. While in Stage II and III, introduction of microaeration resulted in lower effluent COD concentration of 400–800 mg/L and higher COD removal efficiency of 40–70%. It was worth noting that just after the introduction of microaeration, the effluent COD concentration rose sharply for a few days (Day 76–82), with the highest value of 1750 mg/L. Fig. 2(b) shows that during Stage I VFAs in the effluent accounted for about 70% COD and during the Day 76–82 there was not obvious VFAs concentration increasing. So this stream of high COD effluent could mainly contain some biodegradation products of complex organic compounds that had not been degraded to VFAs yet. And this will be further discussed in the next section. In other words, these days played a role of delay period, after which the COD concentration declined quickly to 500 mg/L and maintained steadily below 600 mg/L until the air flux was reduced. With decreasing of COD in effluent, sharp reduction of VFAs was also observed. In petrochemical wastewater pretreatment using hydrolysis-acidification process, Wu et al. (2015) reported that proper limited aeration with dissolved oxygen (DO) under 0.3 mg/L had a positive effect on VFAs production compared to the control reactor, but the production of VFAs decreased significantly when DO was higher than 0.3 mg/L. Similar results were shown during improved anaerobic co-digestion of brown water and food waste by microaeration (Lim et al., 2014). VFAs are easily degradable carbon sources, so the high aeration condition could have further oxidized the generated VFAs (Botheju and Bakke, 2011). Thus the microaeration flux was

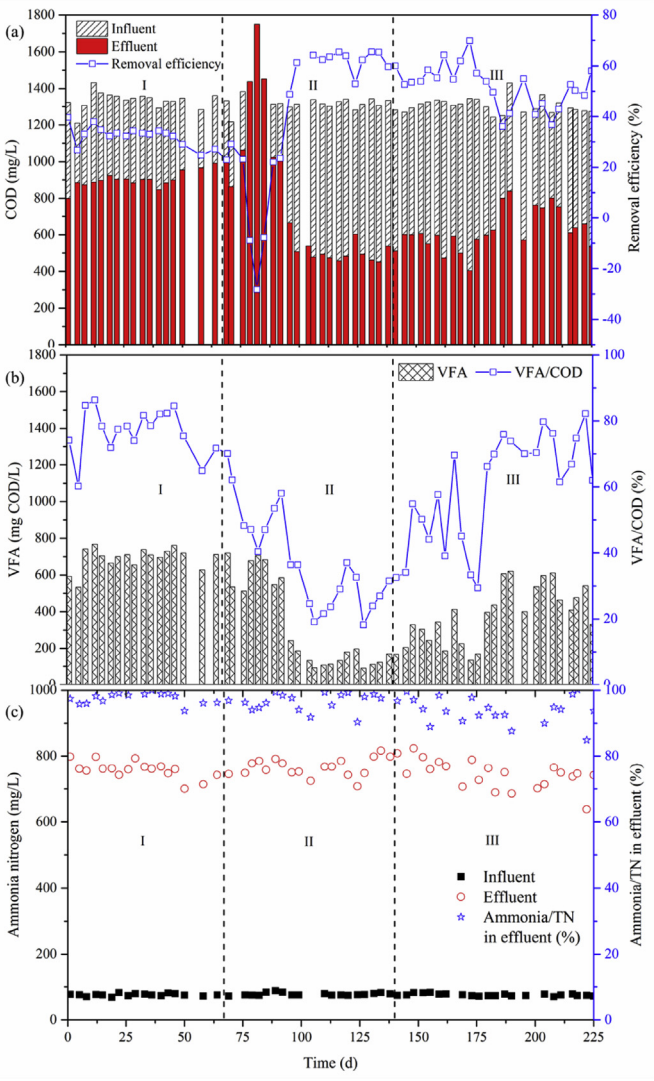


Fig. 2. Performance of the AnMBR on COD removal (a), VFAs production (b) and nitrogen transformation (c).

reduced in Stage III and subsequently the VFAs to COD ratio of the effluent gradually rose back to 60–80%.

The performance of the AnMBR on nitrogen transformation is shown in Fig. 2(c). In the influent organic nitrogen namely urea took the dominant proportion of TN, and the ammonia nitrogen concentration was only 70–80 mg/L. Under the activities of ammonifying bacteria, almost all the urea was transformed into ammonia nitrogen and the ammonia nitrogen concentration of the effluent took a percentage of above 90% of TN, whatever with or without microaeration. This indicated that the oxygen introduced by the microaeration was nearly completely consumed in the hydrolysis-acidogenic process involved with organic carbon compounds biodegradation. The performance of the AnMBR on nitrogen transformation and VFAs production proved that it was running well at the hydrolysis-acidogenic state as designed.

3.2. Accumulation and degradation of surfactants in the AnMBR

Fig. 3(b) shows that the MBAS concentration of influent was about 110 mg/L throughout the whole experimental period. In Stage I the MBAS concentration of effluent was 40–70 mg/L most of

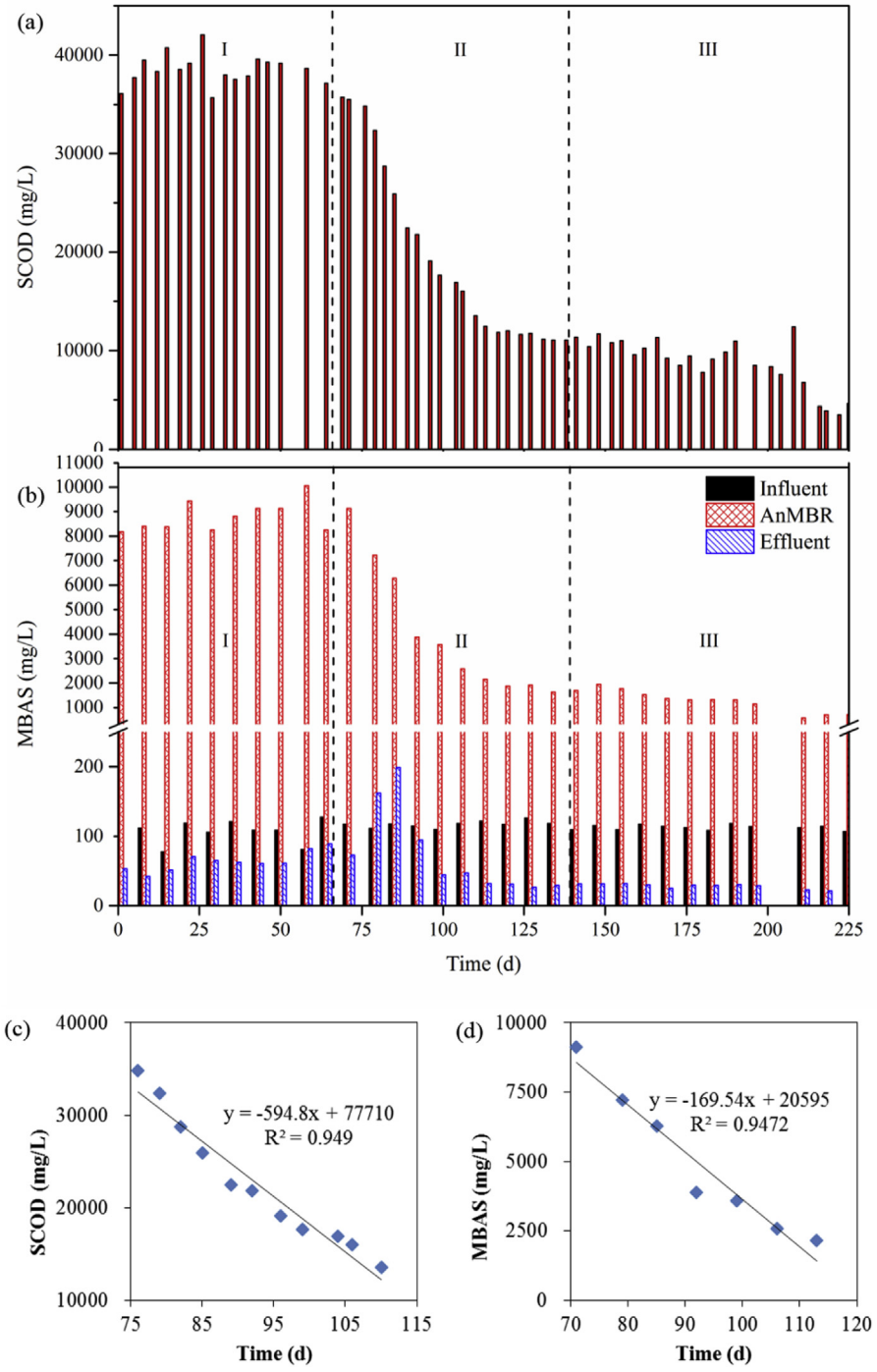


Fig. 3. SCOD of supernatants (a), evolution of surfactants degradation (b) and degradation kinetics of SCOD (c) and surfactants (d) during Stage II.

the time. However it was soon discovered that the reduced surfactants were not actually degraded but retained in the AnMBR by the separation of tubular membrane, which was reflected by the high MBAS concentration of supernatant in the AnMBR even above 10000 mg/L. The molecular weight of SLS and SLES are not too big to make themselves retained by the tubular membrane. When dissolved in water at low concentration surfactant molecules exist as monomers. The concentration of surfactant at which the thermodynamics of the surfactant-solvent system favor micelle formation is called the critical micelle concentration (CMC). At concentrations higher than the CMC surfactants have the ability to solubilize significantly more of a hydrophobic organic compound than would dissolve in water alone (D. Haigh, 1996). That is to say, at concentrations above the CMC, surfactants will connect with more organic matter with their hydrophobic groups and thus format complex compounds whose size are big enough to be intercepted by the membrane separation. The SLS and SLES concentrations in Stage I were obviously higher than their CMC values proposed by Tolls and Sijm (1995) and Tokiwa (1968), respectively. And this resulted in the sustaining accumulation of the surfactants and other organic compounds in the AnMBR contributing to the SCOD concentration as high as 40000 mg/L in the supernatant as

shown in Fig. 3(a).

The accumulation of surfactants and other organic compounds was apparently a serious problem that threatened the steady operation of the AnMBR. As microaeration had been proven effective to enhance the hydrolysis-acidification process (Wu et al., 2015), it was introduced into the AnMBR in this study. The MBAS concentration of the supernatant decreased promptly from 9000 mg/L to about 2000 mg/L when the descending speed began to slow down. And the SCOD concentration of the supernatant showed the same trend of evolution, which confirmed the fact that the high SCOD concentration had been caused by the accumulation of surfactants. Fig. 3(c) and (d) shows that during the first half of Stage II, degradation of surfactants and SCOD followed the kinetic characteristics of zero order reaction with R^2 values of 0.9472 and 0.949. It was worth noting that during the Day 71–85, the MBAS concentration of the effluent was obviously increased at the highest value of 199 mg/L. This was the reason why during these days the effluent COD concentration was increasing sharply (Fig. 2(a)). And in Fig. 3(a) it could be seen that the descending speed of SCOD was relatively slow during the first few days in Stage II. In this period, something might happen like this: the introduced oxygen caused physical-chemical and biological changes in the mixed liquor, and some surfactant molecules were cut from attached organic compounds but large-scale degradation of these separate organic compounds and surfactant molecules had not started yet. So large quantities of surfactant molecules went through the membrane and caused the rising of MBAS and COD concentrations of effluent.

To further investigate the detailed variation caused by the microaeration, 3D-EEM analysis was conducted. The three supernatant samples were collected in different experimental stages. In Fig. 4, three types of fluorophores representing tyrosine-like protein (Region I), tryptophan-like protein (Region II) and soluble microbial by-product-like material (Region III) (Chen et al., 2003) were observed and data within the corresponding three regions were analyzed. It could be seen that without microaeration only fluorophore appeared in Region I. And along with the extension of microaeration, fluorophore showed up in Region III and became stronger. In the meantime the fluorophore in Region I migrated towards Region II. Further more, results of FRI analysis (Table 2) revealed that from the day 43 to day 225 the percent fluorescence response $P_{I,n}$ decreased from 78.5% to 53% while on the other hand $P_{II,n}$ and $P_{III,n}$ increased from 14.9% to 29.7% and 6.6%–17.3%, respectively. As the name suggests, soluble microbial by-product-like material is usually produced by microbial activities. So the significant boost of $P_{III,n}$ indicated that under microaeration life vitality of microbes in the AnMBR was activated and enhanced afterwards, which was responsible for the subsequent reductions of MBAS and SCOD. Amino acid composition is a sensitive indicator for the degradation of organic matter and it has been reported that during organic matter degradation acidic species tyrosine decrease (Wu and Tanoue, 2001). Tryptophan-like fluorescence is related to microbial activity and may be created by microbial activity within the system (autochthonous) (Gone et al., 2009). A linear relationship between the BOD_5 of wastewater and their corresponding fluorescence intensities at 340 nm exists using an excitation wavelength of 280 nm (Reynolds and Ahmad, 1997). These findings were consistent with the evolution of DOM substances indicated by 3D-EEM analysis in this study.

Recently SEC has been used to fractionate organic matter in the environmental engineering processes (Guo et al., 2017). As different molecules with the same molecular weight may have different sizes, the standard substance used for SEC should have the same molecular structure type as the testing samples to acquire accurate results (Kr Egeberg and Alberts, 2003). The DOM in this study had too complex constituents to be determined on the AMW using a known standard substance. But from the SEC results we could still get the AMW distribution information of DOM. In Fig. 5, peaks of effluent samples from different experimental stages appeared in a broad range from 6 to 16min. The large molecules with peak appearing in 6–8 min could be complex micelle-like matter containing surfactants and organic compounds, while medium and small molecules with peak appearing in 12–16 min could represent intermediate metabolites and VFAs. Under microaeration the relative intensity of large molecules decreased and the representing peak nearly disappeared at last whereas there was an obvious increase in the number of small molecules (15–16 min). The variation

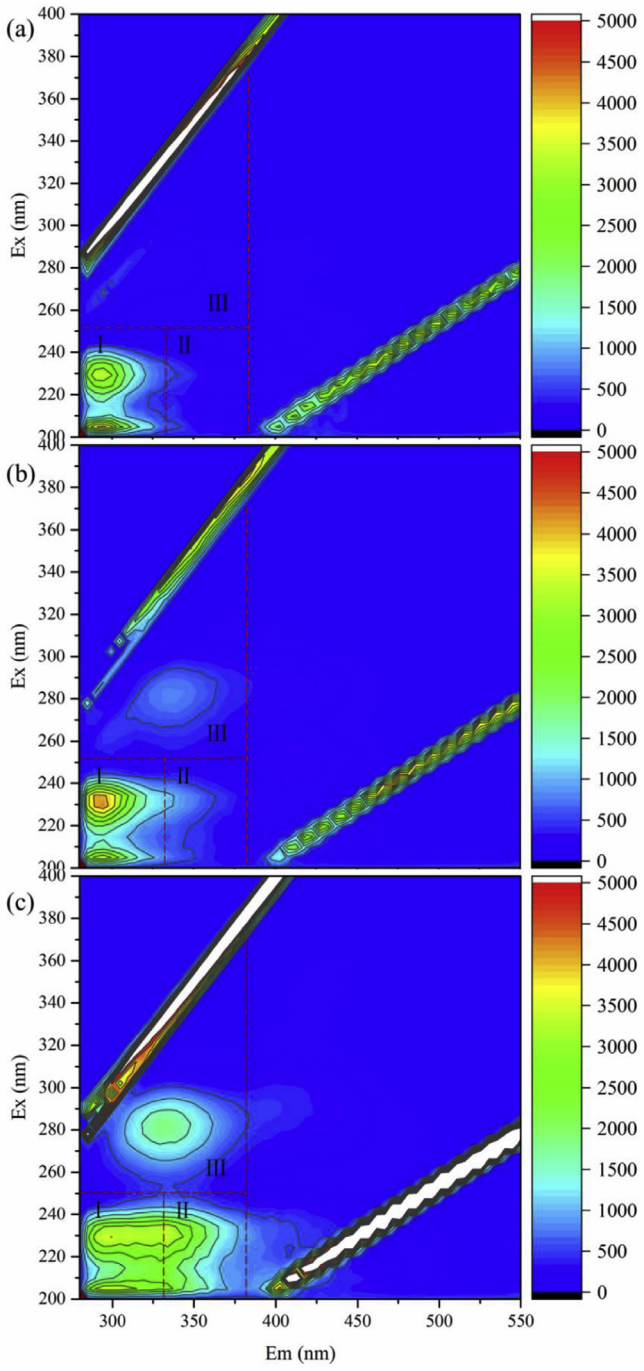


Fig. 4. 3D-EEM fluorescence spectra of filtrates of supernatants on day 43 (a), 82 (b) and 225 (c).

Table 2
FRI parameters and percentage (P) values for EEM analysis of supernatant samples.

FRI parameters					$P_{i,n}$ (%) of samples		
EEM region	No. of EEM data points per region (N)	Projected Ex-Em area (nm^2)	Fractional projected area per region	MF_i	Day 43	Day 82	Day 225
I	121	2500	0.19	5.2	78.5	70.5	53.0
II	110	2500	0.19	5.2	14.9	18.5	29.7
III	294	8000	0.61	1.63	6.6	11.0	17.3
Sum	525	13000	1		100	100	100

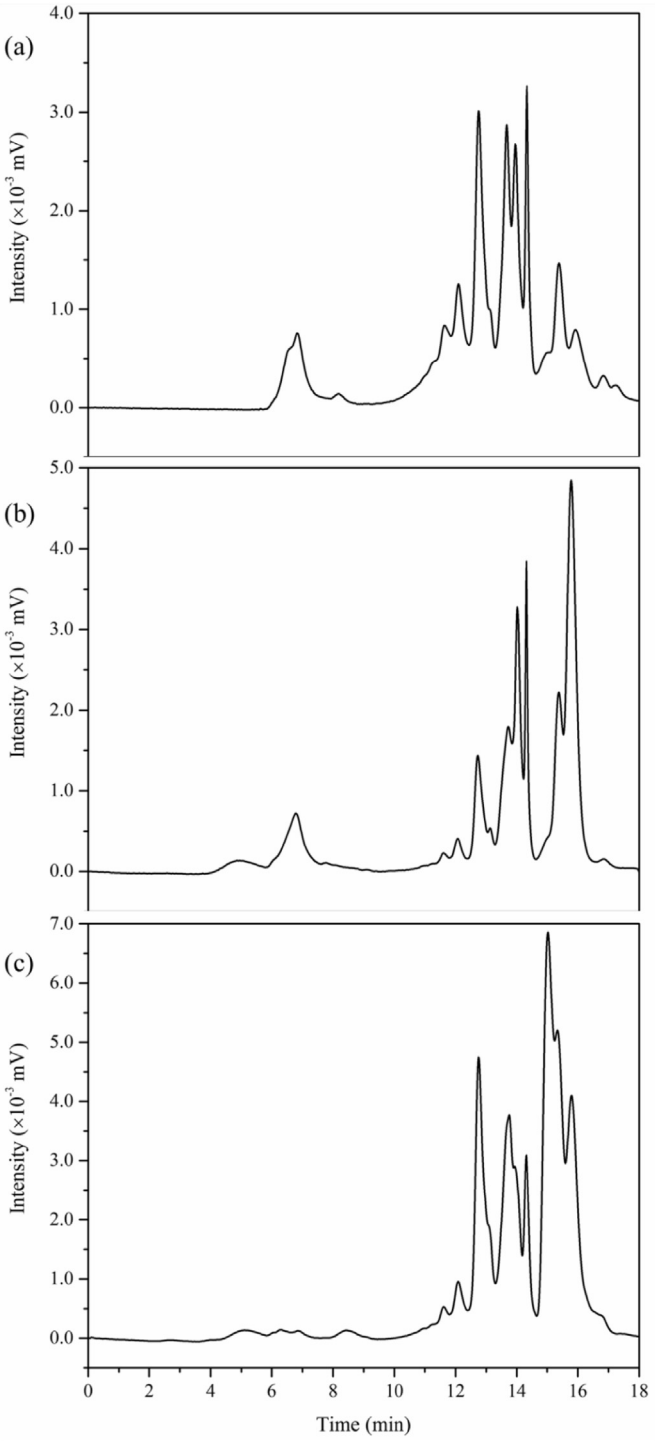


Fig. 5. Apparent molecular weight distribution of the effluent on day 43 (a), 82 (b) and 225 (c).

of AMW distribution further suggested that the complex micelle-like matter that formed and exhibited persistent in the hydrolysis-acidogenic process was disintegrated and then gradually degraded under microaeration condition.

3.3. Microbial community evolution

The high coverage values of OTUs (97% similarity) (Table 3) demonstrated that the sequencing depth was enough to reflect the microorganisms' profile in the inoculum and samples from different experimental stages. The Shannon and Simpson indexes showed that operational conditions exerted significant influence on the microbial community diversity. Under complete anaerobic condition the microbial community diversity was greatly decreased and microaeration had a positive effect on increasing it. But the variation could also have been the result of the continuous declining surfactants concentration.

In Stage I *Firmicutes* and *Proteobacteria* which were two of the three dominant phyla in the inoculum were enhanced whereas *Bacteroidetes* disappeared (Fig. 6 (a)). *Firmicutes* has been reported to play critical roles at hydrolysis and acidification stages (Zheng et al., 2013). And this study indicated that this phylum could resist high concentration of SLS and SLES.

Stage II was a period when the large amounts of accumulated surfactants were biodegraded effectively. Under microaeration condition in this period *Proteobacteria* was further enhanced but the dominant class became *Alphaproteobacteria* instead of *Gammaproteobacteria* (Fig. 6 (b)) and the phylum *Actinobacteria* rose accounting for a percentage of 29%. The emerging genus belonging to *Actinobacteria* was *Flaviflexus* (Fig. 6 (c)) which was proposed by Du et al. (2013) with the type species *Flaviflexus huanghaiensis*. The facultatively anaerobic species was positive for the activity of esterase lipase and could hydrolyze starch and gelatin. Among the dominant class *Alphaproteobacteria Aquamicrobium* was the major genus. All members of the genus *Aquamicrobium* have been isolated from pollutant-loaded environments such as wastewater treatment plants, activated sewage sludge and biofilters (Jin et al., 2013). Some species affiliated to *Aquamicrobium* have been reported to produce varieties of enzymes such as esterase and esterase lipase and thus they were able to degrade complex organic compounds such as petroleum hydrocarbon (Wang et al., 2015) and a wide range of aryloxyphenoxypropionate herbicides (Wang et al., 2017).

In Stage III, low MBAS concentration was maintained in the supernatant while the bacterial community further evolved. The dominant genera became *Thiopseudomonas* and *Pseudomonas*, both

Table 3
Characteristics of microbial community diversity.

Sample	Reads	OTU	Coverage	Shannon	Simpson
Inoculum	23541	1357	0.981	5.05	0.03
Stage I	23844	124	0.997	1.27	0.35
Stage II	23761	251	0.995	2.04	0.25
Stage III	60764	476	0.994	2.54	0.15

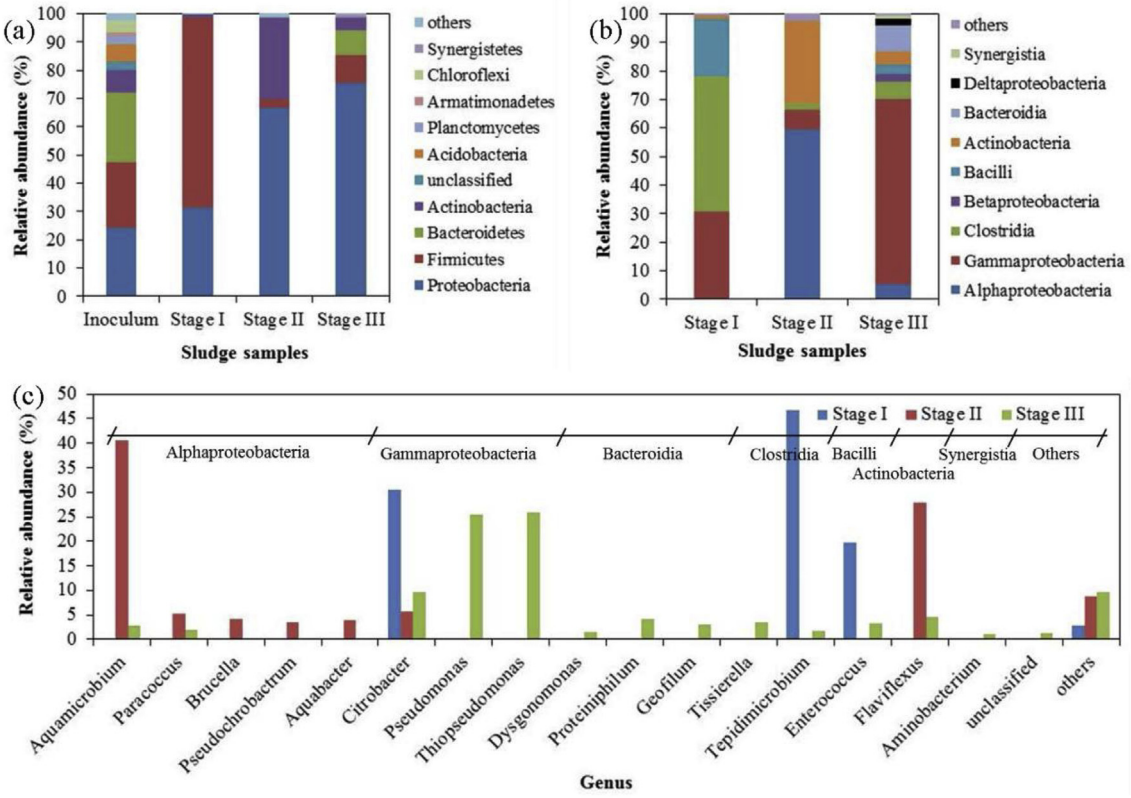


Fig. 6. Relative abundance of the major bacterial community (>1%) at phylum (a), class (b) and genus (c) levels.

taking a share of more than 25%. *Aquamicrobium aestuarii* sp. was reported unable to grow under anaerobic condition (Jin et al., 2013), which could have been the reason why the class *Alphaproteobacteria* significantly decreased when the air flux was reduced in this period. The facultatively anaerobic genus *Thiopseudomonas* was affiliated to the class *Gammaproteobacteria*. It was proposed in 2015 by Tan et al. (2015) with the type species *Thiopseudomonas denitrificans* which was isolated from sludge of an anaerobic, denitrifying, sulfide-removal bioreactor. The genus *Thiopseudomonas* was also positive for activities of esterase and esterase lipase, like *Aquamicrobium* and *Flaviflexus*. The ability to produce these involving enzymes may have been responsible for the degradation of surfactants. Compared with *Thiopseudomonas, Pseudomonas* also affiliated to the class *Gammaproteobacteria* is a quite older genus. It was described in 1894 and is one of the most diverse and ubiquitous bacterial genera whose species have been isolated worldwide in all kinds of environments (Peix et al., 2009). Roig et al. (1998) proved that immobilized *Pseudomonas* C12B was effective SLS degrading biocatalyst and the experimental result proposed by Jovcic et al. (2010) showed that *Pseudomonas* sp. ATCC19151 degraded all SLS present in the liquid medium (up to 4% w/v of SLS) within 48 h at 30 °C. And recently two strains isolated from a MBR that efficiently degraded SLES were confirmed to belong to the genus *Pseudomonas* (Karray et al., 2016).

4. Conclusion

The anionic surfactants SLS and SLES in the influent accumulated in the supernatant of the AnMBR during the hydrolysis-acidogenic process. Introduced microaeration effectively decreased the concentration of surfactants from 9000 mg/L to 2000 mg/L without foam emerging or significant mineralization of VFAs. DOM fractionation results revealed that microaeration enhanced the activities of microbes and led to the disintegration of aggregates of large molecules. Under microaeration condition variation occurred in the bacterial community and the genera *Aquamicrobium*, *Flaviflexus*, *Pseudomonas* and *Thiopseudomonas* might be responsible for the biodegradation of the surfactants. All these results show that microaeration is a valuable approach under hydrolysis-acidogenic condition in CELSS wastewater treatment and reclamation process.

Declarations of interest

None.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.chemosphere.2018.06.001.

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